

Yiwei(Cathy) Wang

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EXPERIENCE

Uber, Software Engineer II | Backend Engineering

2020.01-2024.02

Payment, Risk, Identity Team

- **Payment**.....
- Developed robust **Payment Authentication** solutions, including 2-Factor Authentication (**2FA**), Upfront Charge, Auth Hold, Penny Authorization, and 3-Dimension Secure for credit, debit, and combo card transactions
- Spearheaded **Payments Processing and Settlement** for Uber's Financial Products, managing Uber credits and gift cards via Bank's **Payment Grant** from diverse Bank Card Services such as Ayden and Stripe
- Contributed to the seamless integration of different **Payment Service Providers (PSPs)** such as Amex, Paypal, Google Pay, Apple Pay, and AliPay (HK) into Uber's payment ecosystem
- Integrated a **Standardized Checkout** process with risk check, enhancing Uber's wallet top-up and in-app gifting experience
- **Risk, Identity**.....
- Led initiatives in preventing fraud losses through large-scale real-time **ML training data generation** using **Flink** based streaming feature computation, **Kafka**, **Cassandra**, **Redis**, **DocStore(MySQL-based)** for Promotion Abuse, Refund Fraud, Cancellation Fee Gaming, etc.
- Established a **Global Action Framework** to centralize tiered actioning for verifying drivers' identity and detecting fraudulent behaviors using **Spark** based offline data dispersal, **Piper(Airflow-based)** data pipelines, **Hive**, and **Cassandra**
- Implemented **user-interactive challenges**, including Delivery Pin Verification and Payment Code Verification, for various devices and payment tokens
- **Others**.....
- Successfully migrated a **Microrepo** - a backend service with 55 endpoints for feature fetching - to **Go Monorepo**, enhancing efficiency and scalability
- Conducted **end-to-end testing** and rollout through tenancy, staging, and simulated gateways testing
- Established monitoring metrics with **Grafana** and set up **A/B testing** experiments to ensure smooth deployments

Uber, Software Engineer Intern | ML Backend Engineering

2019.05-2019.08

Marketplace Risk Team, Detection of Invalid Trips in Real-time

- Implemented **ETL**(Extract, Transform, Load) jobs and built **Samza** based pipelines to compute streaming features with near real-time event processing
- Fetched **real-time features** including ML model scores via **RPC** calls and made predictions based on **model-based machine learning** via the decisioning engine
- Designed predicates, fraud actions, and rules for detecting invalid trips using **rule-based machine learning**
- Successfully detected >90% invalid trips, preventing abuse in Uber's driver incentive program

EDUCATION

Cornell University Master of Engineering in Computer Information Science

2018.8-2019.12

- Concentration: Software Engineering
- Relevant Coursework: Object-Oriented Programming, Machine Learning, Database, Data Structures, Operating Systems, Artificial Intelligence, Distributed Computing, Algorithms, Natural Language Processing

SELECTED PROJECTS

Novel 360 Panoramic Video Chat System | Android Developer

2018.08-2019.05

Android App Development

- Developed an Android chat App with panoramic video chat using **Java** and **Android Studio**
- Realized real-time video chatting using **WebRTC** and images with VR view by **Google VR**
- Implemented user authentication with **FireBase** for 3rd party users and multiple log in methods

Visualizing College Pathway Web Application | Front-end Developer

2019.01-2019.05

Web Application

- Developed a web application for course pathways recommendation using **React**, **Django**, **Python**, and **Mongo DB**
- Applied **KNN algorithm** to classify students preference and visualized dynamic data by **D3.js**

SKILLS

- Programming Skills: Java, Go, Python, SQL, JavaScript, Bash, R
- Tools: Spark, Cassandra, DocStore, Flink, Kafka, Hive, Grafana, Samza, Firebase, MongoDB, Android Studio, Django, React, Angular, MySQL, Postgres, Gradle, Git, ElasticSearch, Bootstrap, Jenkins, AWS